

Socioeconomic Differences in Stroke Incidence and Prognosis Under a Universal Healthcare System

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Background and Purpose—Low socioeconomic position (SEP) is associated with high overall stroke mortality, but its contribution to stroke prognosis is unclear. We evaluated socioeconomic disparities in stroke incidence and poststroke outcomes.

Methods—We collected hospital discharge and mortality data for all 35- to 84-year-old Rome residents who had a first acute ischemic or hemorrhagic stroke in 2001 to 2004. We used a small-area SEP index. We calculated age-adjusted incidence rates and rate ratios by SEP for fatal and nonfatal stroke subtypes using Poisson regression. Logistic regression was used to study outcomes by SEP (30-day mortality, and among 1-month survivors: 1-year mortality, hospital readmissions for a successive stroke, and cardiovascular diseases).

Results—A total of 10 033 incident strokes (75% ischemic) were observed. Incidence rates (per 100 000) for ischemic and hemorrhagic stroke were: 104 and 34 in men and 81 and 28 in women, respectively. There were socioeconomic disparities in stroke incidence in both genders (rate ratios between extreme SEP categories in ischemic and hemorrhagic stroke: 1.76; 95% CI, 1.59 to 1.95; 1.50; 95% CI, 1.26 to 1.80 in men; 1.72; 95% CI, 1.55 to 1.91; 1.37; 95% CI, 1.15 to 1.63 in women). No association was found for SEP and mortality after stroke. Men with low SEP with an ischemic event were more likely to be hospitalized for a new stroke than men with high SEP. Women with low SEP with hemorrhagic stroke were more likely to be hospitalized for cardiovascular disease compared with women with high SEP.

Conclusions—Stroke incidence strongly differs between socioeconomic groups reflecting a heterogeneous distribution of lifestyle and clinical risk factors. Strategies for primary prevention should target less affluent people. (*Stroke*. 2009; 40:2812-2819.)

Key Words: hemorrhagic ■ ischemic ■ social class ■ stroke

Stroke is the second leading cause of death in industrialized countries and one of the leading causes of disability worldwide.¹ Stroke incidence is not evenly distributed in the population because several studies have shown a higher incidence of stroke in low socioeconomic groups.²⁻⁴ The social gradient for hemorrhagic stroke tends to be similar to or stronger than that for all stroke,² although a study on Swedish middle-aged women reported greater inequalities for ischemic than for hemorrhagic stroke.³ Despite all of this research, the association between socioeconomic position (SEP) and stroke has been studied much less than socioeconomic inequalities in ischemic heart disease.² Socioeconomic disparities in stroke incidence might be due to a variety of factors. Differential distribution of behavioral or clinical risk factors and of access to health care are all possible explanations for the association between SEP and stroke occurrence as seen for a wide range of other illnesses.² However, some studies have shown that adjusting for several risk factors did not explain the entire association between stroke and socioeconomic position.^{3,5} A recent review of the issue addressed the need for further investigations to evaluate the nature of the association between SEP and stroke incidence.²

The evidence of the association between SEP and survival after stroke is much more controversial because studies conducted in the United States,^{6,7} Finland,⁸ Austria,⁹ and China¹⁰ have all shown a strong relationship between SEP and survival, whereas others have found a weak or even no association at all.² One possible explanation is the differences in provision of care. Theoretically, access to health services in universal healthcare systems (like in Italy where health care is publicly funded and there is universal access and comprehensive coverage under the National Health Service) should be independent of SEP, and the effect of SEP should be less substantial on survival or on other long-term outcomes than on incidence because of the comprehensive and uniform aspect of free health coverage. Studies available on the association between SEP and survival did not show clear differences in diverse healthcare systems.⁹⁻¹¹ In fact, even studies conducted in countries with a universal healthcare system such as Finland or Canada found inequalities in mortality after stroke.^{6,8} Although in the past there was evidence of socioeconomic inequality in outcomes after acute myocardial infarction, new findings from Canada suggest the

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gap has decreased in recent years.¹² Thus, another issue to be considered is that the relationship between SEP and outcomes after stroke could vary over time.

The objective of this study was to analyze the nature of socioeconomic disparities in stroke incidence separately for men and women and by stroke subtype between 2001 and 2004. In addition, we evaluated SEP differences in short-term mortality and in 1-year outcomes (death, rehospitalization for stroke or cardiovascular disease) after stroke. Considering that Italy has a universal healthcare system, which should guarantee equal treatment for the entire population, the latter issue is of particular importance. Given that the distribution of personal lifestyle and clinical risk factors is skewed toward low socioeconomic status, we hypothesized that socioeconomic disparities are present in stroke incidence regardless of gender and stroke subtype, but not in poststroke outcomes, which tend to depend on healthcare quality.

Materials and Methods

Data

This study is based on information from the Health Information System of the Lazio region, where Rome is located. The Regional Cause of Death Registry lists the causes of death coded according to the International Classification of Diseases, 9th Revision, for all deaths of residents of the Lazio region. Discharge abstracts, from both public and private hospitals, are routinely collected by the Regional Information System and contain: patient demographic data (gender, age, place of birth, census block of residence for residents of Rome), admission and discharge dates, up to 6 discharge diagnoses (International Classification of Disease, 9th Revision, Clinical Modification [ICD-9-CM]), medical procedures or surgical interventions (up to 6), and status at discharge (alive, dead, transferred to another hospital). The municipal registry maintains records on all official residents of Rome, including mortality information and date and place of death. All registries use a personal code that allows a subject to be identified in different registries and to access their census block of current and past residences.

Rome has a population of approximately 2.7 millions residents living in 5377 census blocks. Similarly to deprivation indicators used in other countries,¹³ we used a small-area composite index based on 2001 census data (SEP) as a measure of socioeconomic position.¹⁴ In Italy, a census of the entire resident population is conducted every 10 years. For each census block with at least 50 inhabitants (4888 census blocks in Rome with an average population of 500 people), we considered aggregate census information that represents various aspects of deprivation: education, occupation, home ownership, family composition, and citizenship. We performed a factor analysis with varimax rotation on all standardized variables and found that 4 factors explained 84% of the overall variance. The first factor (49.5% of the variance) was characterized by education, occupation, and crowding index; the second (14% of the variance) by citizenship; the third (11.3%) was influenced by family composition; and the last (9.1%) by home ownership. We used an algebraic combination of these factors to create an index of socioeconomic position by census block, distributed in quintiles, ranging from very well off (Level 1) to very underprivileged (Level 5). We attributed the SEP index to individual cases (see subsequently) through census blocks of residence. We observed a correlation of 0.76 with a census block income index we built with 1998 data.¹⁵

Selection of Incident Cases

We selected all hospitalizations and deaths with a diagnosis, or cause of death, of ischemic (International Classification of Diseases, 9th Revision 434, 436) or hemorrhagic stroke (International Classification of Diseases, 9th Revision 430, 431) that occurred from 2001 to 2004 in 35- to 84-year-old Rome residents. We considered hospital

discharge or death from stroke of the same subject that occurred within 28 days of each other as a single stroke event. To detect incident cases, we excluded all hospitalized cases with a secondary diagnosis of late effects of cerebrovascular disease (ICD-9-CM: 438) and those who were discharged from the hospital in the previous 5 years with a diagnosis of stroke or late effects of cerebrovascular disease (ICD-9-CM: 430, 431, 434, 436, 438). To eliminate potential iatrogenic strokes, we also excluded all cases that had heart surgery or a carotid endarterectomy in the previous month or during the index hospital event.¹⁶ For details on ICD-9-CM codes, see Supplemental Table II, available online at <http://stroke.ahajournals.org>.

We considered out-of-hospital deaths (incident cases from the death registry, ie, subjects with a first stroke who died before reaching a hospital), hospitalized cases, and total incidence. Hospital deaths and people who died within 30 days of the event were considered fatal cases.

Clinical Risk Factors for Stroke and Comorbidities

Based on ICD-9-CM codes and following the definitions of a validated coding algorithm (enhanced Elixhauser AHRQ-Web-ICD-9-CM),¹⁷ we identified a priori the clinical risk factors for stroke: hypertension, atrial fibrillation, diabetes, ischemic heart disease, congestive heart failure, and peripheral vascular disorders, because they have commonly been considered in previous studies as confounders or potential mediators in the association between SEP indicators and health outcomes.^{6,9–11,18}

We also identified comorbidities (chronic lung disease, liver disease, renal disease, cancer, coagulation disorders, and anemia) that worsen prognoses in patients affected by cardiovascular disease or other conditions.^{19,20} The differential distribution of these conditions among social classes in specific cardiovascular cohorts based on administrative data has already been noted.^{6,15,21} Information on these conditions was found in the clinical records from the index stroke hospitalization and in the records of all hospitalizations that occurred in the 2 years before the event. For details on ICD-9-CM codes, see Supplemental Table II.

Short-Term and Long-Term Outcomes

Hospitalized cases were followed-up for 1 year. We linked our data set to the Regional Cause of Death Registry to study socioeconomic inequalities in 30-day mortality and, in those who survived 30 days, in 1-year mortality after stroke. We compared our data with discharge abstracts to identify hospital readmissions within 1 year of a first stroke event in survivors after 30 days. We analyzed rehospitalizations for stroke (ICD-9-CM: 430 to 438) and for cardiovascular diseases (ICD-9-CM: 390 to 459) separately for ischemic and hemorrhagic incident cases.

Data Analysis

We used Poisson models to calculate age-adjusted incidence rates (fatal events, nonfatal events, total events) in the 35- to 84-year-old Rome population by stroke subtype, socioeconomic position index, stratified by gender (reported as annual rates per 100 000 persons), and age-adjusted rate ratios with 95% CIs using the highest SEP quintile as the reference category. Denominators were obtained for each year for 5-year age groups, gender, and socioeconomic position index from the Municipal Registry Office database.

Frequencies of clinical risk factors for stroke and comorbidities in the index hospital admission and during the previous 2 years were tabulated for hospitalized cases by area-based socioeconomic position and stroke subtype. Nonparametric tests for trend across ordered groups were calculated.

Given that the follow-up of this study lasted 1 year, logistic regression was used to study socioeconomic disparities in short-term and 1-year outcomes after ischemic and hemorrhagic stroke. Odds ratios (ORs) with 95% CIs were calculated with the highest SEP quintile as the reference category adjusting for age as a continuous variable (an age-squared term was introduced when needed). Clustering within census blocks was taken into account using robust

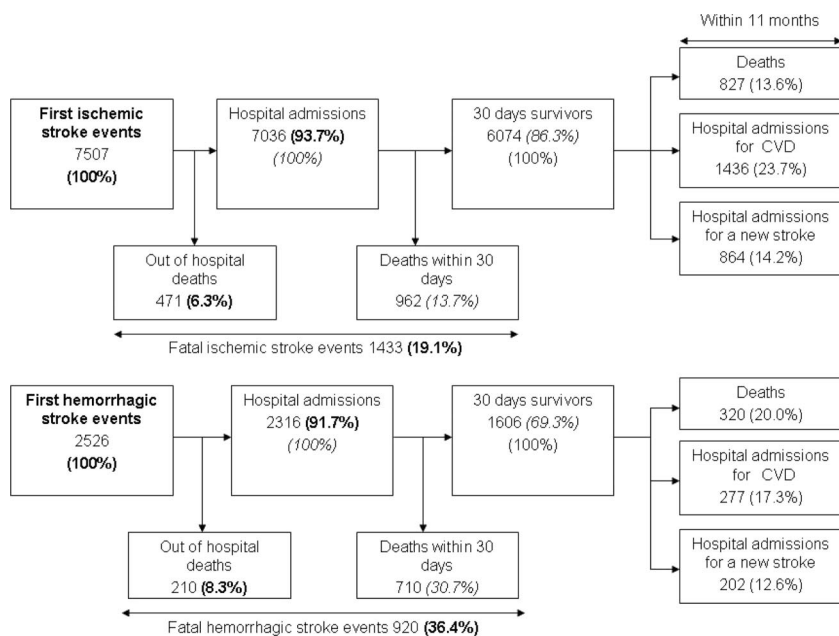


Figure. First ischemic and hemorrhagic stroke and their outcomes in the population of Rome, aged 35 to 84 years, 2001 to 2004.

variance estimation. We used logistic regression instead of a survival analysis because we were interested in evaluating cumulative incidence rather than incidence density of the outcomes 1 year after the stroke.

To evaluate whether the effect of SEP on stroke prognosis was possibly mediated by clinical risk factors, we performed a stratified analysis. Specifically, we divided stroke cases into 2 groups based on the presence of at least one of the following conditions: hypertension and peripheral vascular disorders (only from hospitalizations in the previous 2 years), atrial fibrillation, diabetes, ischemic heart disease, or congestive heart failure (from the index hospitalization or from the previous 2 years). Some diagnoses were considered only from previous hospitalizations to avoid the well-known coding bias effect, ie, the tendency to omit reporting less severe diagnoses in the hospital discharge records (like hypertension) for severely ill patients.²¹ We used the likelihood ratio test to test the interaction between socioeconomic position and the presence of clinical risk factors.

Results

The Figure summarizes the data used to calculate incidence rates and short- and long-term outcomes. Over the 4-year period, in residents aged 35 to 84 years, there were 7507 new cases of ischemic stroke (6% died out of the hospital, 94% were hospitalized) and 2526 new cases of hemorrhagic stroke (8% died out of the hospital, 92% were hospitalized). Fourteen percent of patients admitted to the hospital for an ischemic stroke and 31% of those admitted for a hemorrhagic stroke died within 30 days. Nine patients with an ischemic stroke and 2 with a hemorrhagic stroke moved outside of the study area within 1 month after their hospital admission and were lost to follow-up. Among 30-day ischemic stroke survivors, 14% died within 1 year of the event, 24% were hospitalized for a cardiovascular disease, and 14% were hospitalized for a new stroke event. Among 30-day hemorrhagic stroke survivors, 20% died within 1 year of the event, 17% were hospitalized for a cardiovascular disease, and 13% were hospitalized for another stroke event.

Table 1 reports, separately for ischemic and hemorrhagic stroke, the distribution of total incident cases, out-of-hospital deaths, and hospitalized cases by gender, age, area-based

socioeconomic position, and country of birth. Mean age was 72 years (SD 9.6) at first ischemic stroke and 67.7 years (SD 12) at first hemorrhagic stroke. In both stroke subtypes, out-of-hospital deaths were more common among women.

Table 2 shows, by stroke subtype, gender-specific age-adjusted incidence rates and rate ratios of first stroke by socioeconomic position and case fatality. Both men and women from low socioeconomic groups experienced higher rates of both ischemic and hemorrhagic stroke more frequently than high SEP groups; the strength of the association tended to be higher for ischemic stroke than for hemorrhagic stroke. The socioeconomic gradient was similar for men and women in all cases and nonfatal events, whereas it was strong in men and weak in women in fatal events.

The frequency distributions of clinical risk factors and comorbidities by socioeconomic position among those with ischemic and hemorrhagic stroke are reported in Supplemental Table I, available online at <http://stroke.ahajournals.org>. For ischemic stroke cases, there was an association between socioeconomic position and prevalence of diabetes and atrial fibrillation with higher prevalence in more disadvantaged groups both when measured in the index admission and in the 2 years before the event. Chronic lung diseases (in hospitalizations during the 2 years before the event) were also differently distributed. For hemorrhagic stroke cases, those living in disadvantaged census blocks more frequently had diabetes, liver disease (both when measured in the index admission and in the 2 years before the event), ischemic heart diseases (in the index hospitalization), and renal disease (in the 2 years before the event).

Table 3 shows the association between area-based socioeconomic position and outcomes after the stroke event. The relative frequency of the outcomes (crude risk) and the OR adjusted for age are reported. There was no evidence of socioeconomic disparities in short-term or first-year fatality in either ischemic or hemorrhagic cases. Among males with ischemic stroke, there was an indication of increased risk of

Table 1. Characteristics of Residents in Rome (Aged 35–84 Years) Who Had a First Stroke Event During the Period 2001–2004

	Ischemic Stroke						Hemorrhagic Stroke					
	Total Incidence		Out-of-Hospital Deaths		Hospitalized Cases		Total Incidence		Out-of-Hospital Deaths		Hospitalized Cases	
	No.	Percent	No.	Percent	No.	Percent	No.	Percent	No.	Percent	No.	Percent
Gender												
Men	3944	52.5	220	46.7	3724	52.9	1301	51.5	94	44.8	1207	52.1
Women	3563	47.5	251	53.3	3312	47.1	1225	48.5	116	55.2	1109	47.9
Age, years												
35–44	111	1.5	2	0.4	109	1.6	147	5.8	15	7.1	132	5.7
45–54	374	5.0	3	0.6	371	5.3	269	10.7	19	9.1	250	10.8
55–64	967	12.9	20	4.3	947	13.5	442	17.5	27	12.9	415	17.9
65–74	2370	31.6	81	17.2	2289	32.5	764	30.3	59	28.1	705	30.4
75–84	3685	49.1	365	77.5	3320	47.2	904	35.8	90	42.9	814	35.2
Area-based socioeconomic position												
1 (high)	1236	16.5	99	21.0	1137	16.2	449	17.8	45	21.4	404	17.4
2	1401	18.7	109	23.1	1292	18.4	521	20.6	43	20.5	478	20.6
3	1579	21.0	100	21.2	1479	21.0	472	18.7	39	18.6	433	18.7
4	1572	20.9	86	18.3	1486	21.1	543	21.5	40	19.1	503	21.7
5 (low)	1719	22.9	77	16.4	1642	23.3	541	21.4	43	20.5	498	21.5
Country of birth												
Italy	7255	96.6	451	95.8	6804	96.7	2409	95.4	191	91.0	2218	95.8
Other	252	3.4	20	4.3	232	3.3	117	4.6	19	9.1	98	4.2

cardiovascular disease rehospitalization in the lowest socioeconomic group compared with higher groups and the association became stronger when evaluating the occurrence of a new stroke episode. There was no evidence of an association between socioeconomic position and long-term outcomes in women with ischemic stroke. Among males with hemorrhagic stroke, there was a suggestion of social disparities in rehospitalization, both for stroke and for cardiovascular disease, but the results were not statistically significant. Among women of the lowest socioeconomic group, there was a higher risk of being hospitalized for cardiovascular disease in the year after a hemorrhagic stroke.

We conducted a sensitivity analysis using a Cox proportional model to evaluate the association of SEP and 1-year outcomes after stroke, and the results in terms of hazard ratios generally confirm the findings and provide very similar results, although statistical significance was lost for 1-year rehospitalization for stroke among men with ischemic stroke (hazard ratio, among extreme categories of SEP, 1.30; 95% CI, 0.95 to 1.79) and for 1-year hospitalization for cardiovascular disease in women with hemorrhagic stroke (hazard ratio among extreme categories of SEP, 1.42; 95% CI, 0.81 to 2.47).

Forty-nine percent of ischemic cases and 25% of hemorrhagic cases had clinical risk factors for stroke reported in previous and index hospitalizations. Subjects with clinical risk factors were older than those without these conditions (mean age, 73.3 years; SD 8.4 versus 70.2 years; SD 10.5 for ischemic cases; 71.5 years; SD 9.5 versus 66.2; SD 12.4 for hemorrhagic cases). In general, there was no evidence that socioeconomic position had a different effect on prognosis in the 2 groups (data not shown). The only finding to note is the

degree of the association of SEP with a new stroke in ischemic male cases, which was stronger in subjects without clinical risk factors (OR, 1.61; 95% CI, 1.01 to 2.54 for the lowest SEP category compared with the highest) than among those with clinical risk factors (OR, 1.37; 95% CI, 0.84 to 2.25), although the overall test for interaction was far from being statistically significant ($P=0.953$).

Discussion

We found strong socioeconomic disparities in ischemic and hemorrhagic stroke incidence for both genders, especially for nonfatal events. There was also evidence of socioeconomic differences in incidence of fatal ischemic stroke for men. We did not find evidence of inequalities in 30-day or in 1-year mortality after a first episode of stroke. Nevertheless, there were SEP differences in 1-year readmission rates for stroke in men who had a first ischemic stroke and in 1-year readmission for any cardiovascular disease in women who had a hemorrhagic stroke.

The social gradient we found in the incidence of stroke could be related to clinical and lifestyle risk factors, which are strongly interrelated and are all more common in low socioeconomic groups.²² It is interesting to note that clinical risk factors and comorbidities were not evenly distributed among socioeconomic groups and they were different by subtype. Few studies have shown socioeconomic differences in comorbidities by stroke subtype.^{4,5,7,22} Another possible explanation of our finding could be the socioeconomic differences in use of high-quality healthcare services,^{6,23} but the extent to which preventive and community care (both general and specialized) contributes to disparities in stroke incidence is still debated.⁴

Table 2. Area-Based SEP Index and Incidence of First Stroke in Rome, 2001–2004: Age-Adjusted Rates, Age-Adjusted Rate Ratios, and 95% CIs

SEP	Men				Women			
	No. of Cases	Adjusted Rates	Rate Ratios	95% CI	No. of Cases	Adjusted Rates	Rate Ratios	95% CI
Ischemic stroke								
Fatal events								
1 (high)	118	15.9	1.00		156	17.2	1.00	
2	142	18.9	1.18	0.93–1.51	152	16.6	0.97	0.77–1.21
3	144	19.7	1.24	0.97–1.58	154	17.9	1.04	0.83–1.30
4	149	22.3	1.40	1.10–1.79	129	17.3	1.01	0.80–1.28
5 (low)	158	28.2	1.77	1.40–2.25	133	20.4	1.19	0.95–1.50
Nonfatal events								
1 (high)	502	67.7	1.00		460	50.6	1.00	
2	589	76.7	1.13	1.01–1.28	518	56.1	1.11	0.98–1.26
3	691	91.4	1.35	1.20–1.51	590	67.0	1.32	1.17–1.50
4	700	99.5	1.47	1.31–1.65	595	75.9	1.50	1.33–1.70
5 (low)	752	118.8	1.75	1.57–1.96	676	96.0	1.90	1.69–2.14
Total ischemic stroke events								
1 (high)	620	83.7	1.00		616	67.7	1.00	
2	731	95.6	1.14	1.03–1.27	670	72.7	1.07	0.96–1.20
3	835	111.2	1.33	1.20–1.47	744	84.8	1.25	1.13–1.39
4	848	121.8	1.46	1.31–1.62	724	93.4	1.38	1.24–1.53
5 (low)	910	146.9	1.76	1.59–1.95	809	116.8	1.72	1.55–1.91
Hemorrhagic stroke								
Fatal events								
1 (high)	78	10.5	1.00		103	11.3	1.00	
2	94	12.3	1.17	0.86–1.58	103	11.1	0.98	0.75–1.29
3	80	10.6	1.01	0.74–1.38	85	9.6	0.85	0.64–1.13
4	97	13.8	1.31	0.97–1.77	84	10.6	0.94	0.70–1.25
5 (low)	89	14.1	1.34	0.99–1.81	107	14.9	1.31	1.00–1.72
Nonfatal events								
1 (high)	132	17.8	1.00		136	15.0	1.00	
2	180	23.3	1.31	1.05–1.64	144	15.6	1.04	0.82–1.32
3	168	22.0	1.24	0.98–1.55	139	15.6	1.05	0.83–1.32
4	194	27.0	1.52	1.21–1.89	168	20.8	1.39	1.11–1.74
5 (low)	189	28.5	1.60	1.28–2.00	156	21.1	1.41	1.12–1.78
Total hemorrhagic stroke events								
1 (high)	210	28.3	1.00		239	26.3	1.00	
2	274	35.6	1.26	1.05–1.50	247	26.7	1.02	0.85–1.21
3	248	32.6	1.15	0.96–1.38	224	25.3	0.96	0.80–1.15
4	291	40.8	1.44	1.21–1.72	252	31.4	1.20	1.00–1.43
5 (low)	278	42.6	1.50	1.26–1.80	263	35.9	1.37	1.15–1.63

The lower incidence we found in women is confirmed in the literature.^{7,24} Less obvious is the similar socioeconomic gradient we saw in both genders for ischemic stroke, because socioeconomic inequalities are usually stronger in men than in women.²⁵

Out-of-hospital mortality in women contributes more to total incidence than male out-of-hospital mortality. This is not surprising; gender differences in recognizing symptoms of both myocardial infarction and stroke and late referral to the hospital have been reported among women,^{26,27} and this could

cause a higher probability of dying before hospitalization. It is interesting that we did not find socioeconomic differences in fatal events with the exception of men with ischemic stroke. This could be explained by more serious acute stroke among underprivileged men compared with the well-off male population. Alternatively, it is plausible to hypothesize a delay in healthcare interventions in the acute phase in men or inequalities in access to health care.²

Although the association between SEP and survival after myocardial infarction is well established, there are contradic-

Table 3. Association Between Area-Based SEP and Outcome of First Ischemic and Hemorrhagic Stroke Hospitalization in Rome, 2001–2004

SEP	Men			Women		
	Risk (%)	OR*	95% CI	Risk (%)	OR*	95% CI
Ischemic stroke						
Short-term fatality (30 days)						
1 (high)	13.2	1.00		17.7	1.00	
2	14.3	1.17	0.84–1.63	14.4	0.79	0.58–1.08
3	12.9	1.08	0.78–1.50	14.0	0.79	0.58–1.08
4	12.9	1.18	0.86–1.62	12.6	0.72	0.53–0.99
5 (low)	12.7	1.23	0.90–1.69	13.4	0.82	0.61–1.10
First-year fatality						
1 (high)	13.2			15.0		
2	12.2	0.97	0.67–1.40	17.8	1.24	0.88–1.76
3	12.6	1.06	0.75–1.50	15.1	1.06	0.74–1.52
4	9.6	0.82	0.57–1.18	15.3	1.12	0.79–1.60
5 (low)	11.6	1.10	0.77–1.57	15.9	1.26	0.89–1.78
Rehospitalization for cardiovascular disease (1 year)						
1 (high)	22.3			24.2		
2	25.0	1.17	0.88–1.55	20.3	0.80	0.59–1.08
3	26.5	1.28	0.97–1.67	20.0	0.79	0.59–1.06
4	24.3	1.15	0.87–1.51	25.6	1.09	0.82–1.45
5 (low)	26.6	1.31	1.00–1.72	20.5	0.82	0.62–1.09
Rehospitalization for stroke (1 year)						
1 (high)	11.8			12.9		
2	14.8	1.30	0.91–1.85	10.6	0.81	0.55–1.19
3	17.0	1.53	1.10–2.14	10.7	0.81	0.55–1.19
4	17.0	1.54	1.10–2.15	15.5	1.24	0.87–1.78
5 (low)	16.6	1.50	1.07–2.09	13.1	1.02	0.71–1.46
Hemorrhagic stroke						
Short-term fatality (30 days)						
1 (high)	32.3	1.00		34.9	1.00	
2	29.1	0.86	0.58–1.28	35.7	1.06	0.71–1.57
3	27.9	0.82	0.55–1.23	30.5	0.84	0.56–1.28
4	26.8	0.75	0.50–1.13	29.4	0.80	0.54–1.19
5 (low)	27.3	0.82	0.55–1.23	34.5	1.05	0.70–1.57
First-year fatality						
1 (high)	17.4			20.7		
2	20.6	1.30	0.72–2.35	20.8	1.10	0.61–2.00
3	20.2	1.23	0.69–2.21	24.5	1.38	0.77–2.48
4	25.9	1.59	0.90–2.80	16.1	0.86	0.47–1.57
5 (low)	15.3	0.93	0.51–1.70	18.0	1.15	0.64–2.08
Rehospitalization for cardiovascular disease (1 year)						
1 (high)	13.6			14.1		
2	16.1	1.22	0.65–2.30	18.1	1.32	0.70–2.50
3	19.6	1.55	0.82–2.92	12.2	0.84	0.42–1.69
4	19.2	1.49	0.81–2.72	14.3	1.00	0.52–1.91
5 (low)	19.1	1.52	0.82–2.81	24.4	1.85	1.01–3.40
Rehospitalization for stroke (1 year)						
1 (high)	9.9			11.1		
2	8.9	0.90	0.42–1.92	13.9	1.26	0.62–2.56
3	15.5	1.68	0.82–3.45	9.4	0.81	0.37–1.77
4	15.0	1.61	0.80–3.22	11.9	1.05	0.51–2.15
5 (low)	11.6	1.21	0.59–2.49	18.0	1.61	0.81–3.18

*OR adjusted for age as a continuous variable.

tory results on the inequalities in stroke survival. Some studies have shown a direct association between SEP and survival,^{6,8,10} whereas others have not.^{11,18} Both individual and area-based indices of socioeconomic position were used: education, occupation, income, Castairs scores, and indicators based on census data.² Similar to results from a previous study on survival after myocardial infarction in Rome,²¹ and after selected surgeries,¹⁵ we did not find any association between socioeconomic position and 30-day mortality after stroke. A possible explanation could be that, despite evidence on social inequalities in health status,²⁵ once a subject in the acute phase enters the healthcare system, the health care provided is equitable. Less obvious is that we did not find evidence of socioeconomic disparities in 1-year mortality, in which other factors play a role, different from those involved in acute medical intervention. The other long-term outcomes associated with SEP were readmission for stroke in ischemic incident cases among men and readmission for cardiovascular disease after hemorrhagic stroke among women. This result is consistent with a study by Aslanyan and colleagues who found an association between deprivation and readmission to the hospital for any vascular event after ischemic stroke.¹⁸ The greater need for hospitalization among low socioeconomic groups shows that they are more vulnerable and have a more severe risk profile. Although there was no clear evidence of interaction between SEP and the presence of clinical risk factors, when we examined prognoses among subjects without any apparent clinical risk factors, socioeconomic disparities in 1-year new stroke hospitalizations were found in ischemic cases in men. This result supports the hypothesis that aspects of socioeconomic disparities after stroke include not only individual health status, but also the quality of the healthcare processes. Thus, in addition to primary prevention interventions, it would be advisable to boost long-term care and rehabilitation for underprivileged groups.

Some limitations of the study have to be mentioned. The study is based on administrative data, and despite their wide and valued use as a source for healthcare research, hospital discharge data have several limitations, which have been repeatedly recognized.²⁸ In our case, we did not know clinical and lifestyle risk factors, so we could not directly evaluate differences in blood pressure, smoking habits, diet, and physical activity. To calculate incidence of stroke, we looked back to hospital admissions over the past 5 years to exclude survivors. This procedure could overestimate incident cases, but it is unlikely that this procedure could lead to selection bias. We did not use a larger time window because we think that the quality of discharge data is not very good before 1996. Moreover, we used 2-year discharge abstracts to identify clinical risk factors. As proven by a discharge abstract validation study conducted in the Lazio region, underreporting of secondary diagnoses and procedures is possible.²⁹ However, it is unlikely that different reporting and misclassification errors of comorbidities are associated with SEP. It is more probable, on the contrary, that true incidence of complications and their severity may be higher than reported, weakening the evidence of socioeconomic disparities. Different comorbidities measurements based on ICD-9-CM or ICD-10-CM codes have been tested in cardiovascular research³⁰; the

Elixhauser method allows a comprehensive definition of comorbidities and has proven superior to others in predicting mortality after selected conditions.³¹

We used a small area-based index of socioeconomic position. The use of area-based indices as a proxy of individual traits could lead to a misclassification of individual socioeconomic position.¹³ Moreover, this approach has some limitations; it is impossible to disentangle what is causing the effect, and for its construction, some bias can be introduced. The SEP index includes citizenship, which could reflect other factors beyond socioeconomics (ie, genetic susceptibility). On the other hand, this could also reflect characteristics of the area that are not captured using an individual index. However, because the census blocks in Rome are rather small (a few hundred inhabitants), the misclassification, if any, is likely to be minor. Nevertheless, a similar indicator based on the 1991 census in Rome was associated with many health outcomes.^{21,25} It would be useful to study disabilities and handicaps, but unfortunately, the data were not available for this study.

In conclusion, we found evidence of socioeconomic disparities in stroke incidence and in its risk profiles. The results of this study show that SEP does not have a strong impact on outcome of treatment after a first stroke, in the acute phase, or after 1 year, suggesting that the universal coverage health system is able to mitigate socioeconomic disparities with hospital care. Tackling health inequalities in stroke should focus on interventions, particularly on primary prevention in underprivileged groups of the population.

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Disclosures

None.

References

1. World Health Organization. *World Health Report 2004 Changing History*. Geneva: WHO; 2004 (Statistical Annex).
2. Cox AM, McKeivitt C, Rudd AG, Wolfe CD. Socioeconomic status and stroke. *Lancet Neurol*. 2006;5:181–188.
3. Kuper H, Adami HO, Theorell T, Weiderpass E. The socioeconomic gradient in the incidence of stroke: a prospective study in middle-aged women in Sweden. *Stroke*. 2007;38:27–33.
4. Avendano M, Boshuizen HC, Schellevis FG, Mackenbach JP, Van Lenthe FJ, Van den Bos GA. Disparities in stroke preventive care in general practice did not explain socioeconomic disparities in stroke. *J Clin Epidemiol*. 2006;59:1285–1294.
5. Brown P, Guy M, Broad J. Individual socio-economic status, community socio-economic status and stroke in New Zealand: a case control study. *Soc Sci Med*. 2005;61:1174–1188.
6. Kapral MK, Wang H, Mamdani M, Tu JV. Effect of socioeconomic status on treatment and mortality after stroke. *Stroke*. 2002;33:268–273.
7. Gillum RF, Mussolino ME. Education, poverty, and stroke incidence in whites and blacks: the NHANES I Epidemiologic Follow-up Study. *J Clin Epidemiol*. 2003;56:188–195.
8. Jakovljevic D, Sarti C, Sivenius J, Torppa J, Mahonen M, Immonen-Raiha P, Kaarsalo E, Alhainen K, Kuulasmaa K, Tuomilehto J, Puska P, Salomaa V. Socioeconomic status and ischemic stroke: the FINMONICA Stroke Register. *Stroke*. 2001;32:1492–1498.

9. Arrich J, Lalouschek W, Mullner M. Influence of socioeconomic status on mortality after stroke: retrospective cohort study. *Stroke*. 2005;36:310–314.
10. Zhou G, Liu X, Xu G, Liu X, Zhang R, Zhu W. The effect of socioeconomic status on three-year mortality after first-ever ischemic stroke in Nanjing, China. *BMC Public Health*. 2006;6:227.
11. Weir NU, Gunkel A, McDowall M, Dennis MS. Study of the relationship between social deprivation and outcome after stroke. *Stroke*. 2005;36:815–819.
12. Pilote L, Tu JV, Humphries K, Behouli H, Belisle P, Austin PC, Joseph L. Socioeconomic status, access to health care, and outcomes after acute myocardial infarction in Canada's universal health care system. *Med Care*. 2007;45:638–646.
13. Galobardes B, Shaw M, Lawlor DA, Lynch JW, Davey Smith G. Indicators of socioeconomic position (part 2). *J Epidemiol Community Health*. 2006;60:95–101.
14. Cesaroni G, Agabiti N, Rosati R, Forastiere F, Perucci CA. An index of socioeconomic position based on 2001 census, Rome. *Epidemiol Prev*. 2006;30:352–357.
15. Agabiti N, Cesaroni G, Picciotto S, Bisanti L, Caranci N, Costa G, Forastiere F, Marinacci C, Pandolfi P, Russo A, Perucci CA. The association of socioeconomic disadvantage with postoperative complications after major elective cardiovascular surgery. *J Epidemiol Community Health*. 2008;62:882–889.
16. Skjelland M, Krohg-Sørensen K, Tennøe B, Bakke SJ, Brucher R, Russell D. Cerebral microemboli and brain injury during carotid artery endarterectomy and stenting. *Stroke*. 2009;40:230–234.
17. Quan H, Sundararajan V, Halfon P, Fong A, Burnand B, Luthi JC, Saunders LD, Beck CA, Feasby TE, Ghali WA. Coding algorithms for defining comorbidities in ICD-9-CM and ICD-10 administrative data. *Med Care*. 2005;43:1130–1139.
18. Aslanyan S, Weir CJ, Lees KR, Reid JL, McInnes GT. Effect of area-based deprivation on the severity, subtype, and outcome of ischemic stroke. *Stroke*. 2003;34:2623–2628.
19. Mehta RH, Honeycutt E, Shaw LK, Milano CA, Smith PK, Harrington RA, Sketch MH Jr. Clinical and angiographic correlates of short- and long-term mortality in patients undergoing coronary artery bypass grafting. *Am J Cardiol*. 2007;100:1538–1542.
20. Johnston JA, Wagner DP, Timmons S, Welsh D, Tsevat J, Render ML. Impact of different measures of comorbid disease on predicted mortality of intensive care unit patients. *Med Care*. 2002;40:929–940.
21. Picciotto S, Forastiere F, Stafoggia M, D'Ippoliti D, Ancona C, Perucci CA. Associations of area based deprivation status and individual educational attainment with incidence, treatment, and prognosis of first coronary event in Rome, Italy. *J Epidemiol Community Health*. 2006;60:37–43.
22. Avendano M, Kawachi I, Van Lenthe F, Boshuizen HC, Mackenbach JP, Van den Bos GA, Fay ME, Berkman LF. Socioeconomic status and stroke incidence in the US elderly: the role of risk factors in the EPESE study. *Stroke*. 2006;37:1368–1373.
23. McKevitt C, Coshall C, Tilling K, Wolfe C. Are there inequalities in the provision of stroke care? Analysis of an inner-city stroke register. *Stroke*. 2005;36:315–320.
24. Pessah-Rasmussen H, Engstrom G, Jerntorp I, Janzon L. Increasing stroke incidence and decreasing case fatality, 1989–1998: a study from the stroke register in Malmo, Sweden. *Stroke*. 2003;34:913–918.
25. Cesaroni G, Agabiti N, Forastiere F, Ancona C, Perucci CA. Socioeconomic differentials in premature mortality in Rome: changes from 1990 to 2001. *BMC Public Health*. 2006;6:270.
26. Guru V, Fremes SE, Austin PC, Blackstone EH, Tu JV. Gender differences in outcomes after hospital discharge from coronary artery bypass grafting. *Circulation*. 2006;113:507–516.
27. Ayala C, Croft JB, Greenlund KJ, Keenan NL, Donehoo RS, Malarcher AM, Mensah GA. Sex differences in US mortality rates for stroke and stroke subtypes by race/ethnicity and age, 1995–1998. *Stroke*. 2002;33:1197–1201.
28. Iezzoni LI. Assessing quality using administrative data. *Ann Intern Med*. 1997;127:666–674.
29. Agency for Public Health, Lazio Region (Italy). La valutazione della qualità della compilazione e codifica della scheda di dimissione. Available at: www.asplazio.it/asp_online/att_ospedaliera/sio_newr/sio_altre_publicazioni.php?argo=02&sio=altrepublic. Accessed May 15, 2009.
30. Baldwin LM, Klabunde CN, Green P, Barlow W, Wright G. In search of the perfect comorbidity measure for use with administrative claims data: does it exist? *Med Care*. 2006;44:745–753.
31. Southern DA, Quan H, Ghali WA. Comparison of the Elixhauser and Charlson/Deyo methods of comorbidity measurement in administrative data. *Med Care*. 2004;42:355–360.